

Impact of broad-spectrum antibiotic exposures and multidrug-resistant gram-negative bacteremia on hematopoietic cell transplantation outcomes

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Funding information

Indian Council of Medical Research, Grant/Award Number: IRIS ID 2020-0664

Abstract

Introduction: There is a close association between the use of broad-spectrum antibiotics, gut microbiome alteration, multidrug resistant (MDR) gram-negative bacilli (GNB) bacteremia, graft versus host disease (GVHD), and mortality post-allogeneic hematopoietic cell transplantation (allo-HCT). This study reports the impact of the high use of carbapenems and colistin and MDR bacteremia pre- and post-HCT on HCT outcomes.

Methods: This was a single-center, partial retrospective, and prospective study from 2016 to 2020. Both pre- and post-HCT antibiotic exposures and blood culture/sensitivity were recorded. MDR GNB was defined as either non-susceptibility to third-generation cephalosporin or carbapenems. In the absence of positive cultures, the treating physician escalated antibiotics from third-generation cephalosporins to carbapenem and/or colistin as per clinical discretion. De-escalation policy was not strictly enforced.

Results: MDR GNB bacteremia was seen in 29 of 76 (38%) of patients peri-HCT. The utilization rates for carbapenems and colistin was significantly higher in the cohort with MDR GNB bacteremia pre-HCT (70% vs. 32%, $p = 0.002$ and 31% vs. 6.4%, $p = 0.007$, respectively) and post-HCT (100% vs. 74.5%, $p = 0.002$, and 55.2% vs. 8.5%, $p < 0.0001$, respectively). The cohort with MDR GNB bacteremia had significantly more severe acute GVHD at day+100 (45% vs. 17.5%, $p = 0.009$). The median survival was 204 days compared to not reached in the cohort without any MDR GNB bacteremia ($p = 0.005$).

Conclusion: This study shows pre- and post-HCT MDR GNB bacteremia is associated with an increased risk of severe acute GVHD and mortality. Patients with MDR GNB bacteremia had higher exposure to pre- and post-HCT carbapenems and colistin.

Abbreviations: CRE, carbapenem-resistant Enterobacterales; ESBL, extended-spectrum β -lactamase; GNB, gram-negative bacilli; GVHD, graft versus host disease; HCT, hematopoietic cell transplantation; MDR, multidrug resistance

KEYWORDS

broad-spectrum antibiotics, GVHD, HCT, MDR GNB bacteremia

1 | INTRODUCTION

Febrile neutropenia is almost universal in patients undergoing hematopoietic cell transplantation (HCT). The Infectious Diseases Society of America (IDSA) provided clinical practice guidelines for the use of antimicrobial agents in febrile neutropenia in 2011.¹ The choice of antibiotics is dictated by the prevalence of drug resistance in the local environment and previous infection or colonization with the resistant organisms. However, this is increasingly being challenged in the era of gut microbiome studies showing an adverse impact of antibiotics on HCT outcomes.^{2,3} These studies show that the use of antibiotics reduces gut microbial diversity and increases the risk of gram-negative bacteremia, impacting HCT outcomes like acute graft versus host disease (GVHD) and mortality.^{4,5} The incidence of extended-spectrum β -lactamase (ESBL)-producing gram-negative bacteria and carbapenemase-producing organisms, including carbapenem-resistant Enterobacterales (CRE), varies by geographic region (10%–40%)⁶ and is on the rise in Indian HCT centers.^{7,8} The treatment of ESBL and AmpC β -lactamase producing organisms involves carbapenems, and CRE treatment consists of novel β -lactam/ β -lactamase inhibitors or polymyxin/colistin.⁹ One reason for the increased occurrence of drug-resistant bacteremia is the higher use of antibiotics in the pre-HCT period.¹⁰ Though several studies have investigated the incidence of drug-resistant organisms on fecal surveillance,^{8,11} few have investigated the impact of broad-spectrum antibiotics and drug-resistant bacteremia before the HCT. Here, we study the impact of peri-transplant broad-spectrum antibiotics and drug-resistant gram-negative bacilli (GNB) bacteremia on HCT outcomes.

2 | METHODS

This was a single-center, retrospective study from 2016 to 2018 and prospective from 2019 to 2020. All consecutive patients of >12 years of age who underwent allogeneic (allo)-HCT were included after informed consent. All HCTs were done in high-efficiency particulate air filtered rooms. Any pre-HCT use of antibiotics, and blood culture sensitivity reports during prior febrile neutropenia periods were recorded. Patients receiving myeloablative conditioning (MAC), not reduced-intensity conditioning (RIC), were given levofloxacin prophylaxis 750 mg once daily till onset of febrile neutropenia or neutrophil engraftment.¹² The use of antibiotic post-HCT and blood culture/sensitivity reports during the conditioning and post-transplant neutropenic phases were documented. The first, second-, and third-line antibiotics at our center included cefoperazone-sulbactam, imipenem, and colistin, respectively, in a patient without prior isolation of

resistant organisms. These are based on our institute's antibiotic susceptibility profile of GNB blood isolates to third-generation cephalosporins, carbapenems, and colistin being in the range of 20%–30%, 40%–60%, and ~100%, respectively. The escalation of antibiotics was as per transplant physician discretion in the absence of positive cultures. The antibiotics were modified as per susceptibility in the event of isolation of resistant organisms. The antibiotics were continued until engraftment or defervescence, whichever was later. There was no infectious disease specialist to strictly enforce antibiotic de-escalation policy. An automated BACTEC 9240 (BD Becton Dickinson, USA) system was used for all blood cultures. Gram stain smears were prepared from beep-positive bottles and sub-cultured. All differentiated colonies obtained after overnight incubation at 37°C were identified by matrix-assisted laser desorption ionization-time of flight mass spectrometry (Microflex LT, Bruker Daltonics, GmbH, Germany) using Biotyper 3.0 (Bruker Daltonics). Drug resistance was defined as isolation of any bacteria with *in vitro* non-susceptibility to at least one antimicrobial agent. While the definition of multidrug resistance (MDR) involves resistance to three or more antimicrobial classes, we have used this term if they were resistant to either third-generation cephalosporins/carbapenems.¹³ Non-susceptibility to third-generation cephalosporins was used as a proxy for ESBL producing organisms, while non-susceptibility to carbapenems was used as a proxy for CRE. The isolation of drug-resistant non-Enterobacterales like *Pseudomonas*, *Acinetobacter*, intrinsically resistant non-fermenting GNB like *Burkholderia*, *Stenotrophomonas* was also recorded. All the breakpoints for interpretation were as per the Clinical and Laboratory Standards Institute guidelines (M100 Ed 24) using disk diffusion or Vitek 2 automated system (BioMerieux). Overall survival was defined as the time from the day of hematopoietic cell infusion to death from any cause. The incidence of relapse and acute GVHD was also documented. The severity of acute GVHD was graded as per modified CIBMTR criteria.¹⁴ Statistical analysis was done with Graphpad Prism version 5.0. Categorical variables were compared using chi-square or Fisher exact test. Continuous variables were compared using the independent *t*-test for normally distributed and the Mann-Whitney test for skewed variables. Survival was estimated using the Kaplan-Meier method and differences compared with the log-rank test. Statistical significance was ascertained at a level of *p*-value < 0.05.

3 | RESULTS

Data of 76 allo-HCT recipients were available during the study period. Of these, a total of 29 (38.1%) patients had MDR GNB bacteremia peri-HCT. Sixteen (21%) patients had it pre-HCT. Of the 17 patients (22.4%) that had it post-HCT, 16 (94%) patients had *de novo* MDR organisms

TABLE 1 Comparison of clinical characteristics and HCT outcomes in patients with and without MDR GNB bacteremia

Characteristics	MDR GNB bacteremia N = 29 ^a N (%)	No MDR GNB bacteremia N = 47 N (%)	p value
Age, (years), median (IQR)	25 (18–38)	23 (16–40)	0.5
Sex, male: female	20:9	34:13	0.8
Diagnosis			
Leukemia	25 (86%)	38 (81%)	0.8
Aplastic anemia	4 (14%)	9 (19%)	
Pre-HCT carbapenem	20 (70%)	15 (32%)	0.002
Pre-HCT colistin	10 (34.5%)	3 (6.4%)	0.003
MAC	16 (55%)	27 (57%)	1.0
RIC/NMA	13 (45%)	20 (43%)	
MSD	18 (62%)	27 (57%)	0.8
Alternative	11 (38%)	20 (43%)	
PTCy	8 (28%)	16 (34%)	0.6
Post-HCT carbapenem	29 (100%)	35 (74.5%)	0.002
Post-HCT colistin	16 (55.2%)	4 (8.5%)	<0.0001
Neutrophil engraftment (days), median (IQR)	15 (12.5–18.5)	14 (12–18)	0.5
Platelet engraftment (days), median (IQR)	14 (13–25)	12 (11.7–14)	0.004
Hospital stay (days), median (IQR)	30 (20–52)	21 (17–35)	0.04
Severe aGVHD in survivors	12/23 (52.2%)	9/44 (20.4%)	0.01
Severe gut aGVHD in survivors	11/23 (47.8%)	8/44 (18.2%)	0.02
100-day severe acute GVHD incidence	45%	17.5%	0.009
Relapse	5 (17.2%)	9 (19.1%)	1.0
1-year relapse incidence	25%	19.5%	0.6
Deaths	19 (65.5%)	17 (36.2%)	0.01
Cause of death ^b			
Sepsis	12 (41.4%)	6 (12.7%)	0.006
GVHD	6 (20.7%)	6 (12.7%)	0.5
Progressive disease	4 (13.8%)	7 (14.9%)	1.0
RRT	3 (10.4%)	1 (2.1%)	0.1
Deaths within 100 days	12 (41.2%)	3 (6.4%)	0.0006
Median survival	204 days	Not reached	0.005

^a16 pre-HCT, 17 post-HCT, 16 de-novo post-HCT.^bSome patients have overlapping causes of death.

without any prior isolation. The data of organisms isolated pre- and post-HCT are shown in supplementary Table 1. Of the 29 patients in the MDR cohort, there were a total of 38 isolations pre-HCT, 30 GNB (79%), and eight (21%) gram-positive bacteria. There were 10 pan-drug sensitive GNB, seven ESBL, six CR, seven non-fermenting GNB, and one MRSA. The same cohort had 35 isolations post-HCT, 28 GNB (80%), and seven gram-positive organisms. Five were pan-drug sensitive, five were ESBL, 10 were CR, and eight were non-fermenting GNB, and three methicillin-resistant. There were also three pan-drug resistant organisms isolated post-HCT. In contrast, the cohort without MDR had only 17 organisms isolated pre-HCT, 11 GNB and six gram-positive organisms and five post-HCT with three GNB and two gram-positive. There was no difference in the patient characteristics of age, sex, diag-

nosis, HCT conditioning intensity (levofloxacin prophylaxis with MAC), degree of HLA match, or the use of post-transplant cyclophosphamide as GVHD prophylaxis between the cohorts with and without MDR GNB bacteremia (Table 1). However, the utilization rates for carbapenems and colistin were significantly higher in the cohort with MDR GNB bacteremia both pre-HCT (70% vs. 32%, $p = 0.002$ and 34.5% vs. 6.4%, $p = 0.007$, respectively) and post-HCT (100% vs. 74.5%, $p = 0.002$, and 55.2% vs. 8.5%, $p < 0.0001$, respectively) as compared to the cohort without MDR GNB bacteremia. The use of carbapenems and colistin in the absence of proven resistant organisms was seen in 54% and 73% of patients pre- and post-HCT, respectively.

Impact on HCT outcomes: There was no difference in the days to neutrophil engraftment in the two cohorts (15 vs. 14 days, $p = 0.5$).

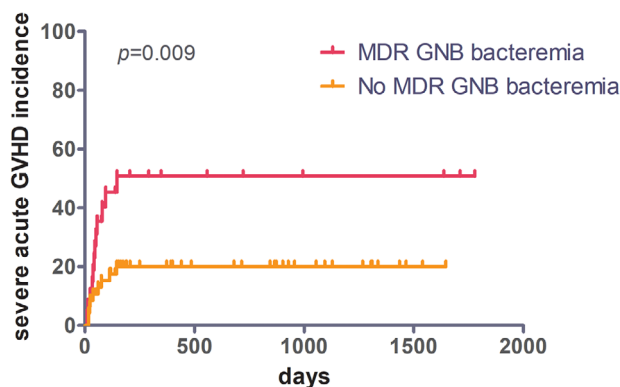


FIGURE 1 Cumulative incidence of severe acute GVHD in the cohorts with and without MDR GNB bacteremia

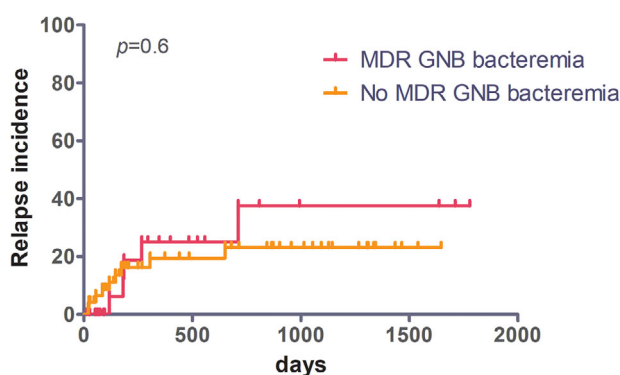


FIGURE 2 Cumulative relapse incidence in the cohorts with and without MDR GNB bacteremia

However, there was a significant median delay of 2 days in platelet engraftment in the cohort with MDR GNB bacteremia (14 vs. 12 days, $p = 0.004$). The duration of hospital stay was significantly longer in the cohort with MDR GNB bacteremia (30 vs. 21 days, $p = 0.04$). Of the survivors without transplant-related mortality, there were significantly more patients with severe acute GVHD (52.2% vs. 20.4%, $p = 0.01$) and severe acute gut GVHD (47.8% vs. 18.2%, $p = 0.02$) in the cohort with MDR GNB bacteremia. The 100-day severe acute GVHD incidence was significantly higher in the cohort with MDR GNB bacteremia (Figure 1) (45% vs. 17.5%, $p = 0.009$). There was no difference in the proportion of patients with relapse (17.2% vs. 19.1%, $p = 1.0$) or the 1-year incidence of relapse (25% vs. 19.5%, $p = 0.6$) (Figure 2). There were a significantly higher proportion of total deaths (65.5% vs. 36.2%, $p = 0.01$) and deaths at 100 days post-HCT (41.2% vs. 6.4%, $p = 0.0006$) in the cohort with MDR GNB bacteremia. The median survival was 204 days compared to not reached in the cohort without any MDR GNB bacteremia ($p = 0.005$) (Figure 3). Significantly more patients died due to bacterial sepsis in the cohort with MDR GNB bacteremia (41.4% vs. 12.7%, $p = 0.006$). There was no difference in GVHD as the cause of death in the two cohorts (20.7% vs. 12.7%, $p = 0.2$).

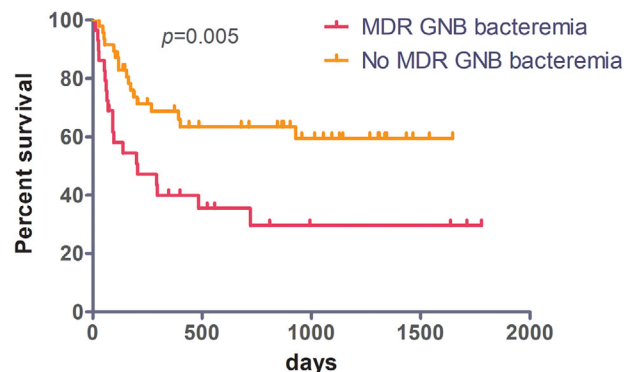


FIGURE 3 Overall survival in the cohorts with and without MDR GNB bacteremia

4 | DISCUSSION

The challenge of drug-resistant organisms is vital to low-middle income countries such as India, with reports of 57.7% colonization by MDR organisms,⁸ 68.8% CRE isolates,⁷ and a high incidence in the community.¹⁵ Our study had MDR GNB bacteremia documented in 38% of patients with an equal number of patients pre- and post-HCT. Gram-positive organisms were of less concern accounting for only 20% of isolates. The utilization rates of carbapenems and colistin, an antibiotic of last resort for MDR GNB infections, were also significantly higher in the cohort with MDR GNB bacteremia pre- and post-HCT. More than half of the patients were exposed to carbapenems and colistin without evidence for a resistant organism. Such pre-HCT exposures, as observed in more than half of the patients in our study, may select for *de novo* drug resistance.¹⁰ This may lead to pan-drug resistance, as was seen in three of our patients. We could not identify any patient characteristics associated with the isolation of MDR GNB organisms, possibly due to small patient numbers. In contrast, previous studies isolating organisms with the same resistance pattern pre- and post-HCT have identified leukemia and mismatched HCT associated with GNB bacteremia.¹⁶ There seems to be no protective effect of levofloxacin prophylaxis in preventing GNB. There were an equal proportion of patients who received it as prophylaxis during MAC in both cohorts. The benefit of fluoroquinolone prophylaxis is increasingly being questioned in the current era.^{3,16} The association of MDR GNB bacteremia with delay in platelet engraftment may be explained due to sepsis-induced thrombocytopenia.¹⁷ The occurrence of MDR GNB bacteremia also prolonged hospital stay.¹⁸

This study shows that pre-HCT and post-HCT MDR GNB bacteremia are associated with an increased risk of severe acute GVHD and mortality. We report a high incidence of severe acute GVHD (45%) in patients with MDR GNB bacteremia, similar to that reported in studies looking at the association of gut colonization by antibiotic-resistant bacteria.¹⁹ We also report very high mortality rates of 40% in patients with MDR GNB bacteremia, similar to those reported in other studies.^{8,20} Several previous studies confirm the association of antibiotics on gut microbiome alterations leading to MDR bacteremia and GVHD and mortality in high-income countries.²⁻⁵ These studies

have demonstrated the negative impact of fluoroquinolone prophylaxis and broad-spectrum antibiotics like carbapenems post-HCT. Just carbapenem exposure, not colonization or MDR bacteremia, increases mortality from 13% to 21.5%.⁵

In a recent EBMT Infectious Disease Working Party survey of centers across Asia and Europe, empirical escalation to broad-spectrum antibiotics was done in 71.4% of centers. Only half of them de-escalated and discontinued antibiotics before neutrophil recovery.²¹ This may be due to uncertainty in the various guidelines, with IDSA guidelines recommending continuing empiric antibiotics until neutrophil recovery¹ and ECIL guidelines recommending discontinuation after 72 h in stable patients.²² Most centers in India do not have infectious disease or antimicrobial stewardship guidance for clinicians to enforce these difficult decisions. The limitations of our study are a single-center, partial retrospective nature, and the use of surrogate definitions for ESBL/CRE/MDR organisms. Given the alarming rise in the incidence of MDR GNB in Indian centers, it is vital that transplant physicians or antimicrobial stewardship programs regulate the use of these broad-spectrum antibiotics. Otherwise, the emergence of pan drug resistance will threaten the field of HCT in India.

ACKNOWLEDGMENTS

Part of this work was sponsored by the Indian Council of Medical Research (ICMR) IRIS ID 2020-0664 grant to DPL. SK did this work on Junior Research Fellowship from the Department of Biotechnology, Government of India.

CONFLICT OF INTEREST

All authors have no conflict of interest to disclose.

AUTHOR CONTRIBUTIONS

Shaweta Kaundal, Pankaj Malhotra, and Deepesh P. Lad conceived the study. Shaweta Kaundal, Harmandeep Singh, Aditya Jandial, Kripa Shanker Kasudhan, Madhu Chopra, Niranjana Khair, and Deepesh P. Lad collected, analyzed the data, and drafted the manuscript. All authors were involved in patient care, reviewed the manuscript, and approved the final version. Shaweta Kaundal, Aditya Jandial, and Deepesh P. Lad confirm full access to the study's data and final responsibility for the manuscript.

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REFERENCES

- Freifeld AG, Bow EJ, Sepkowitz KA, et al. Clinical practice guideline for the use of antimicrobial agents in neutropenic patients with cancer: 2010 Update by the Infectious Diseases Society of America. *Clin Infect Dis*. 2011;52(4):427–431.
- Weber D, Hiergeist A, Weber M, et al. Detrimental effect of broad-spectrum antibiotics on intestinal microbiome diversity in patients after allogeneic stem cell transplantation: lack of commensal sparing antibiotics. *Clin Infect Dis*. 2019;68(8):1303–1310.
- Taur Y, Xavier JB, Lipuma L, et al. Intestinal domination and the risk of bacteremia in patients undergoing allogeneic hematopoietic stem cell transplantation. *Clin Infect Dis*. 2012;55(7):905–914.
- Peled JU, Gomes ALC, Devlin SM, et al. Microbiota as predictor of mortality in allogeneic hematopoietic-cell transplantation. *N Engl J Med*. 2020;382(9):822–834.
- Shono Y, Docampo MD, Peled JU, et al. Increased GVHD-related mortality with broad-spectrum antibiotic use after allogeneic hematopoietic stem cell transplantation in human patients and mice. *Sci Transl Med*. 2016;8(339):339ra371.
- Satlin MJ, Walsh TJ. Multidrug-resistant Enterobacteriaceae, *Pseudomonas aeruginosa*, and vancomycin-resistant *Enterococcus*: three major threats to hematopoietic stem cell transplant recipients. *Transpl Infect Dis*. 2017;19(6):e12762.
- Barman P, Choudhary D, Chopra S, Thukral T. Blood stream infections in hematopoietic stem cell transplant patients: a 2-year study from India. *Oncol J India*. 2020;4(2):43–48.
- Korula A, Perumalla S, Devasia AJ, et al. Drug-resistant organisms are common in fecal surveillance cultures, predict bacteremia and correlate with poorer outcomes in patients undergoing allogeneic stem cell transplants. *Transpl Infect Dis*. 2020;22(3):e13273.
- Satlin MJ, Weissman SJ, Carpenter PA, Seo SK, Shelburne SA. American Society of Transplantation and Cellular Therapy Series, 1: enterobacteriales infection prevention and management after hematopoietic cell transplantation. *Transplant Cell Ther*. 2021;27(2):108–114.
- Averbuch D, Tridello G, Hoek J, et al. Antimicrobial resistance in gram-negative rods causing bacteremia in hematopoietic stem cell transplant recipients: intercontinental prospective study of the infectious diseases working party of the European Bone Marrow Transplantation Group. *Clin Infect Dis*. 2017;65(11):1819–1828.
- Alevizakos M, Karanika S, Detsis M, Mylonakis E. Colonisation with extended-spectrum beta-lactamase-producing Enterobacteriaceae and risk for infection among patients with solid or haematological malignancy: a systematic review and meta-analysis. *Int J Antimicrob Agents*. 2016;48(6):647–654.
- Tomblyn M, Chiller T, Einsele H, et al. Guidelines for preventing infectious complications among hematopoietic cell transplantation recipients: a global perspective. *Biol Blood Marrow Transplant*. 2009;15(10):1143–1238.
- Magiorakos AP, Srinivasan A, Carey RB, et al. Multidrug-resistant, extensively drug-resistant and pandrug-resistant bacteria: an international expert proposal for interim standard definitions for acquired resistance. *Clin Microbiol Infect*. 2012;18(3):268–281.
- Schoemans HM, Lee SJ, Ferrara JL, et al. EBMT-NIH-CIBMTR Task Force position statement on standardized terminology & guidance for graft-versus-host disease assessment. *Bone Marrow Transplant*. 2018;53(11):1401–1415.
- Thacker N, Pereira N, Banavali S, et al. Alarming prevalence of community-acquired multidrug-resistant organisms colonization in children with cancer and implications for therapy: a prospective study. *Indian J Cancer*. 2014;51(4):442–446.
- Girmenia C, Bertaina A, Piciocchi A, et al. Incidence, risk factors and outcome of pre-engraftment gram-negative bacteremia after allogeneic and autologous hematopoietic stem cell transplantation: an Italian Prospective Multicenter Survey. *Clin Infect Dis*. 2017;65(11):1884–1896.
- Venkata C, Kashyap R, Farmer JC, Afessa B. Thrombocytopenia in adult patients with sepsis: incidence, risk factors, and its association with clinical outcome. *J Intensive Care*. 2013;1(1):9.
- Lad DP, Malhotra P, Khadwal A, et al. Infections are a major cause of prolongation of hospital stay in hematopoietic stem cell transplants in tropical, developing countries: profile of a transplant center in North India. *Am Soc Hematol*. 2014;124:5853.
- Bilinski J, Robak K, Peric Z, et al. Impact of gut colonization by antibiotic-resistant bacteria on the outcomes of allogeneic hematopoietic stem cell transplantation: a retrospective, single-center study. *Biol Blood Marrow Transplant*. 2016;22(6):1087–1093.

20. Patriarca F, Cigana C, Massimo D, et al. Risk factors and outcomes of infections by multidrug-resistant gram-negative bacteria in patients undergoing hematopoietic stem cell transplantation. *Biol Blood Marrow Transplant*. 2017;23(2):333–339.
21. Verlinden A, Mikulska M, Knelange NS, et al. Current antimicrobial practice in febrile neutropenia across Europe and Asia: the EBMT Infectious Disease Working Party survey. *Bone Marrow Transplant*. 2020;55(8):1588–1594.
22. Averbuch D, Orasch C, Cordonnier C, et al. European guidelines for empirical antibacterial therapy for febrile neutropenic patients in the era of growing resistance: summary of the 2011 4th European Conference on Infections in Leukemia. *Haematologica*. 2013;98(12):1826–1835.

SUPPORTING INFORMATION

Additional supporting information may be found in the online version of the article at the publisher's website.

How to cite this article: Kaundal S, Jandial A, Singh H, et al. Impact of broad-spectrum antibiotic exposures and multidrug-resistant gram-negative bacteremia on hematopoietic cell transplantation outcomes. *Transpl Infect Dis*. 2021;23:e13717. <https://doi.org/10.1111/tid.13717>